New Wheels Project

Introduction to SQL

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Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

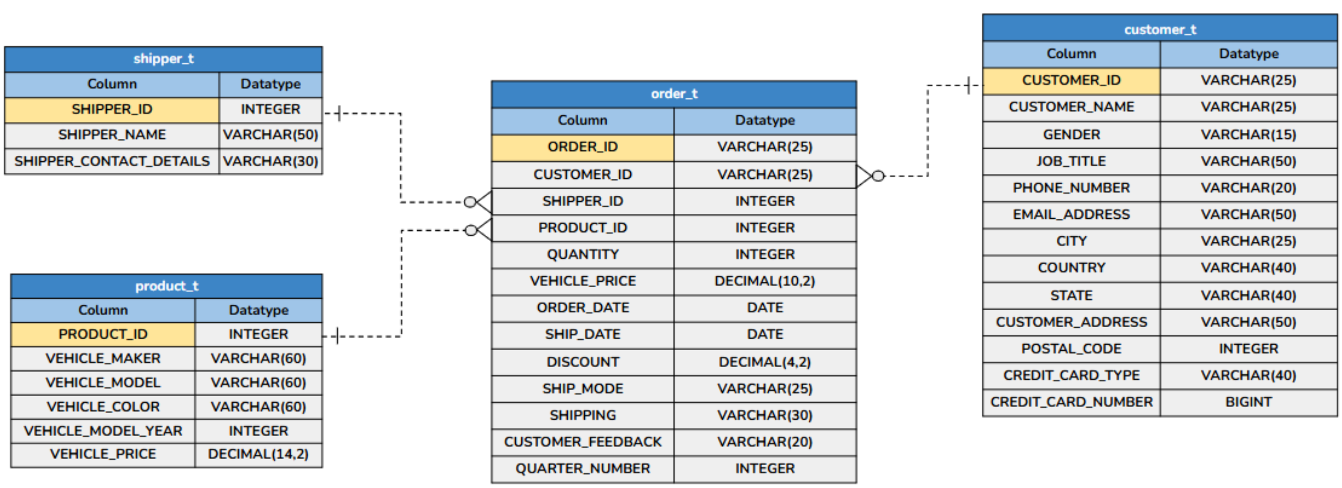
Data Description

The New Wheel database includes the following tables:

- Shipper: Contains details about the shipper, including contact information and name.
- Product: Stores information about products, their pricing, and their model or color.
- Customer: Holds customer information, including contact details and location.
- Orders: Records information about customer orders, including order dates and total amounts, shipping.
- Below are the data columns used in the tables:
  - **shipper_id:** Unique ID of the Shipper
  - **shipper_name:** Name of the Shipper
  - **shipper_contact_details:** Contact detail of the Shipper
  - **product_id:** Unique ID of the Product
  - **vehicle_maker:** Vehicle Manufacturing company name

- **vehicle_model:** Vehicle model name
- **vehicle_color:** Color of the Vehicle
- **vehicle_model_year:** Year of Manufacturing
- **vehicle_price:** Price of the Vehicle
- **quantity:** Ordered Quantity
- **customer_id:** Unique ID of the customer
- **customer_name:** Name of the customer
- **gender:** Gender of the customer
- **job_title:** Job Title of the customer
- **phone_number:** Contact detail of the customer
- **email_address:** Email address of the customer
- **city**: Residing city of the customer
- **country**: Residing country of the customer
- **state:** Residing state of the customer
- **customer_address:** Address of the customer
- **order_date:** Date on which customer ordered the vehicle
- **order_id:** Unique ID of the order
- **ship_date:** Shipment Date
- **ship_mode:** Shipping Mode/Class
- **shipping:** Shipping Ways
- **postal_code:** Postal Code of the customer
- **discount:** Discount given to the customer for the particular order by credit card in percentage
- **credit_card_type:** Credit Card Type
- **credit_card_number:** Credit card number
- **customer_feedback:** Feedback of the customer
- **quarter_number :** Quarter Number

ER Diagram



Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```sql
select count(distinct customer_id) as Total_customer from order_t;

select count(customer_id) as customer_across_state, state from customer_t  group by state order by customer_across_state desc;
```

Output:

Result: Passed

Query 1

Query:

select count(distinct customer_id) as Total_customer from order_t

Output:

Showing 1 rows

Total_customer
994

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

state	customer_across_state
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26

Result 142

Output

Action Output

#	Time	Action	Message
320	22:05:16	select state , count(customer_id) as customer_across_state from customer_t group by state order by customer_across_state LIMIT 0, 10000	49 row(s) returned
321	22:06:04	select state , count(customer_id) as customer_across_state from customer_t group by state order by customer_across_state desc LIMIT 0, 10000	49 row(s) returned

Observations and Insights:

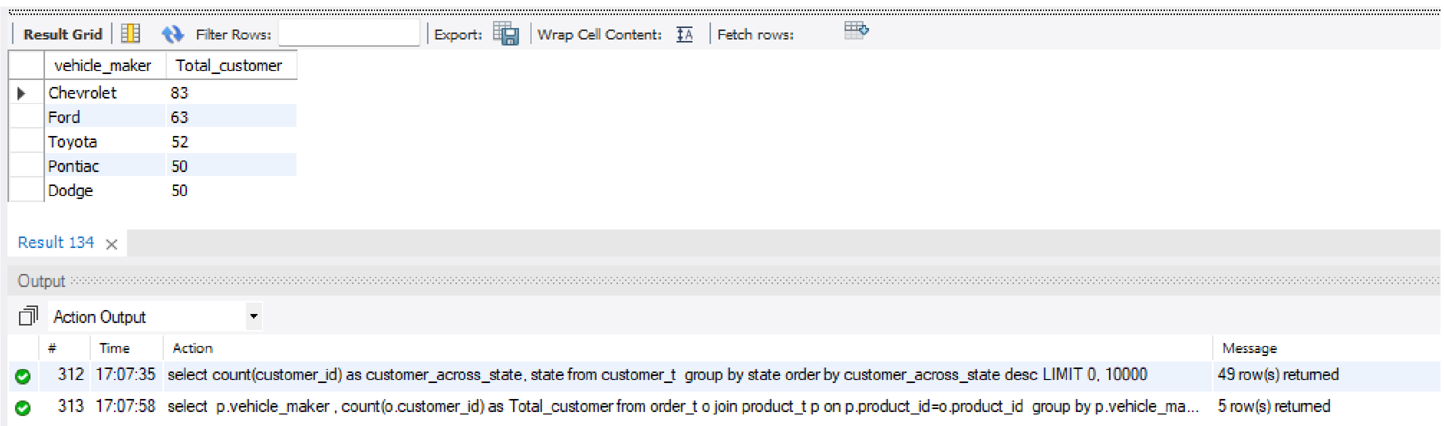
- A total of 994 unique customers have placed orders on the New-Wheels platform, spread across multiple U.S. states.
- California (97) and Texas (97) have the highest number of customers, jointly accounting for nearly 20% of the total customer base, showing strong market penetration in these regions.
- The lowest customer presence is observed in Vermont, Maine, and Wyoming, each with only 1 customer, indicating minimal or no brand penetration.
- Overall, the customer distribution shows high concentration in a few key markets (California, Texas, Florida, and New York) and weak penetration across smaller or rural states.
- To achieve balanced growth, leadership could consider localized marketing campaigns, improved delivery coverage, and strategic reseller tie-ups in low-engagement states.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
select p.vehicle_maker , count(o.customer_id) as Total_customer
from order_t o join product_t p on p.product_id=o.product_id
group by p.vehicle_maker order by Total_customer desc limit 5;
```

Output:



Observations and Insights:

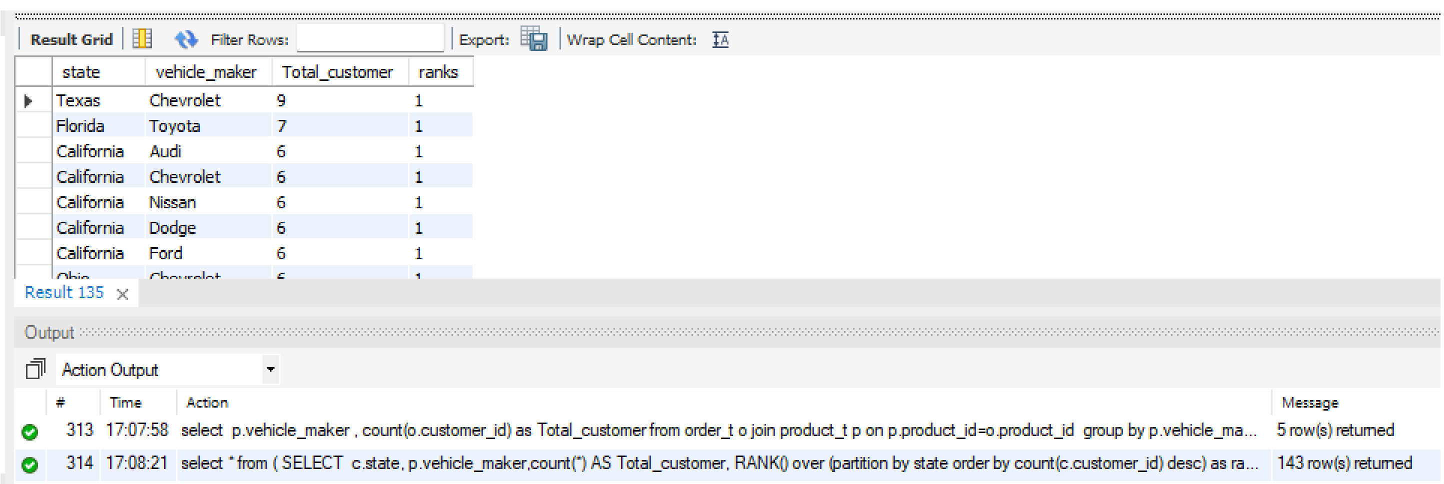
- Chevrolet is the most preferred vehicle maker with 83 customers, followed by Ford (63) and Toyota (52), indicating strong customer trust in brands known for reliability and value retention.
- Pontiac and Dodge, each with 50 customers, show steady mid-tier demand, suggesting loyal niche audiences and potential for targeted growth strategies.
- The data highlights brand familiarity, after-sales service quality, and perceived reliability as key factors influencing customer purchase behavior in the resale market.
- Focusing marketing efforts, partnerships, and inventory planning around these top brands can help maximize sales, improve customer satisfaction, and optimize stock alignment with demand.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
select * from (
SELECT
	c.state,
	p.vehicle_maker,count(*) AS Total_customer,
	RANK() over (partition by state order by count(c.customer_id) desc) as ranks
FROM order_t o JOIN product_t p on p.product_id=o.product_id
JOIN customer_t c on c.customer_id = o.customer_id
GROUP BY c.state,p.vehicle_maker
) preferred_vehicle
where ranks =1
order by Total_customer desc;
```

Output:



state	vehicle_maker	Total_customer	ranks
Texas	Chevrolet	9	1
Florida	Toyota	7	1
California	Audi	6	1
California	Chevrolet	6	1
California	Nissan	6	1
California	Dodge	6	1
California	Ford	6	1

Observations and Insights:

- Chevrolet emerges as the most dominant brand, ranking 1 in multiple states such as Texas, Colorado, Washington, Missouri, Illinois, and the District of Columbia, reflecting strong national appeal and resale trust.
- Toyota shows significant regional popularity, especially in Florida, New York, Georgia, Pennsylvania, and Oregon, highlighting its wide acceptance and reputation for reliability across diverse markets.

- Some states, including California, exhibit diverse brand preferences (Audi, Nissan, Chevrolet, Dodge, and Ford each leading with similar counts), suggesting a competitive market with varied consumer tastes.
- The overall distribution reveals that no single manufacturer dominates all regions so preferences vary by geography, indicating opportunities for region-specific marketing, dealership partnerships, and localized service campaigns to strengthen brand engagement.

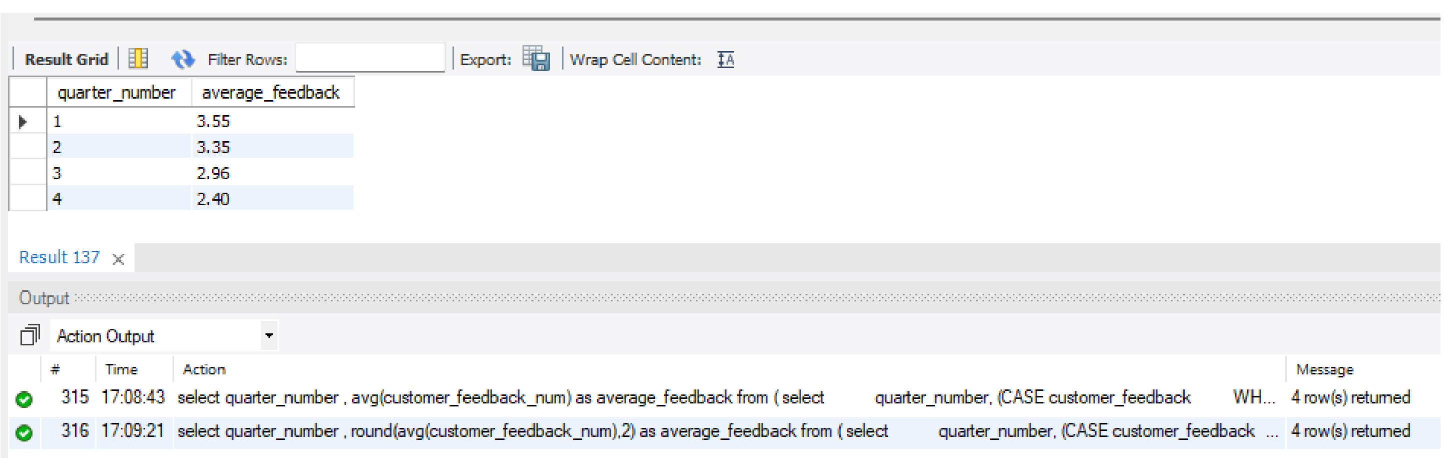
Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
select quarter_number , round(avg(customer_feedback_num),2) as average_feedback
from (

select

        quarter_number, (CASE customer_feedback

        WHEN 'Very Bad' THEN 1

        WHEN 'Bad' THEN 2

        WHEN 'Okay' THEN 3

        WHEN 'Good' THEN 4

        WHEN 'Very Good' THEN 5

    END) AS customer_feedback_num

    FROM order_t

 ) order_tb

 group by quarter_number order by quarter_number;
```

Output:



Observations and Insights:

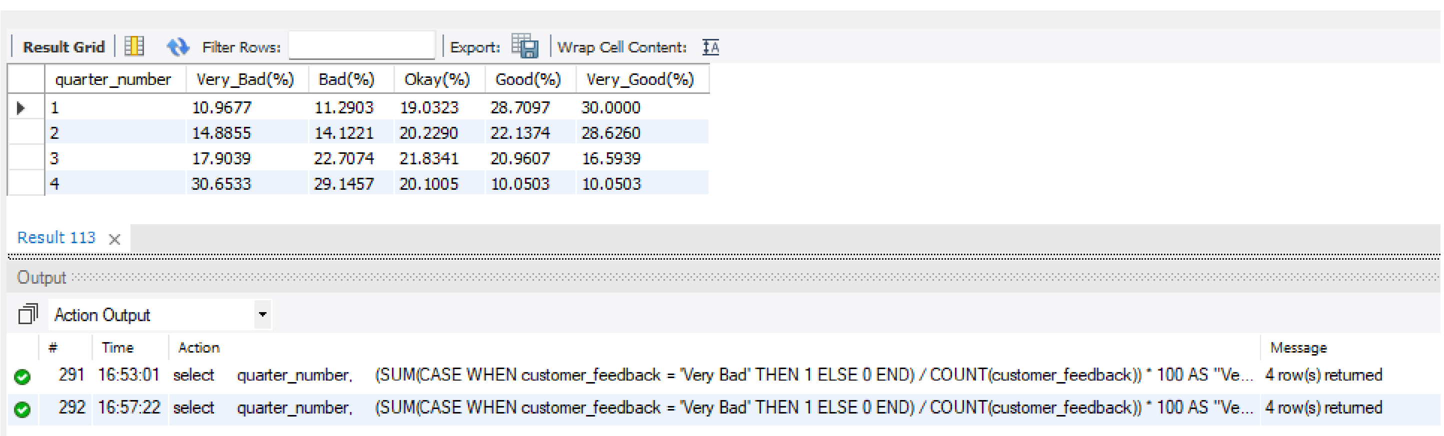
- The overall customer satisfaction shows a declining trend across the year, dropping from an average rating of 3.55 in Q1 to 2.40 in Q4.
- Q1 (3.55) reflects the highest satisfaction level, where most customers rated their experience as Good to Very Good.
- Q4 records the lowest average rating (2.40), signaling a critical drop in customer experience that could be impacting repeat purchases and new customer acquisition.
- The consistent downward pattern suggests the need for root-cause analysis and corrective actions in after-sales service, delivery efficiency, and customer support to rebuild customer trust and improve future feedback scores.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
select     quarter_number,

     (sum(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) /
count(customer_feedback)) * 100 AS "Very_Bad(%)",

     (sum(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) /
count(customer_feedback)) * 100 AS "Bad(%)",

     (sum(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) /
count(customer_feedback)) * 100 AS "Okay(%)",

     (sum(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) /
count(customer_feedback)) * 100 AS "Good(%)",

     (sum(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) /
count(customer_feedback)) * 100 AS "Very_Good(%)"

from order_t

group by quarter_number

order by quarter_number;
```

Output:



quarter_number	Very_Bad(%)	Bad(%)	Okay(%)	Good(%)	Very_Good(%)
1	10.9677	11.2903	19.0323	28.7097	30.0000
2	14.8855	14.1221	20.2290	22.1374	28.6260
3	17.9039	22.7074	21.8341	20.9607	16.5939
4	30.6533	29.1457	20.1005	10.0503	10.0503

Observations and Insights:

- Customer satisfaction has declined consistently across quarters, with positive feedback ('Good' + 'Very Good') dropping from 58.7% in Q1 to just 20% in Q4.
- Meanwhile, negative feedback ('Bad' + 'Very Bad') has sharply risen from 22.3% in Q1 to nearly 60% in Q4, indicating growing dissatisfaction.

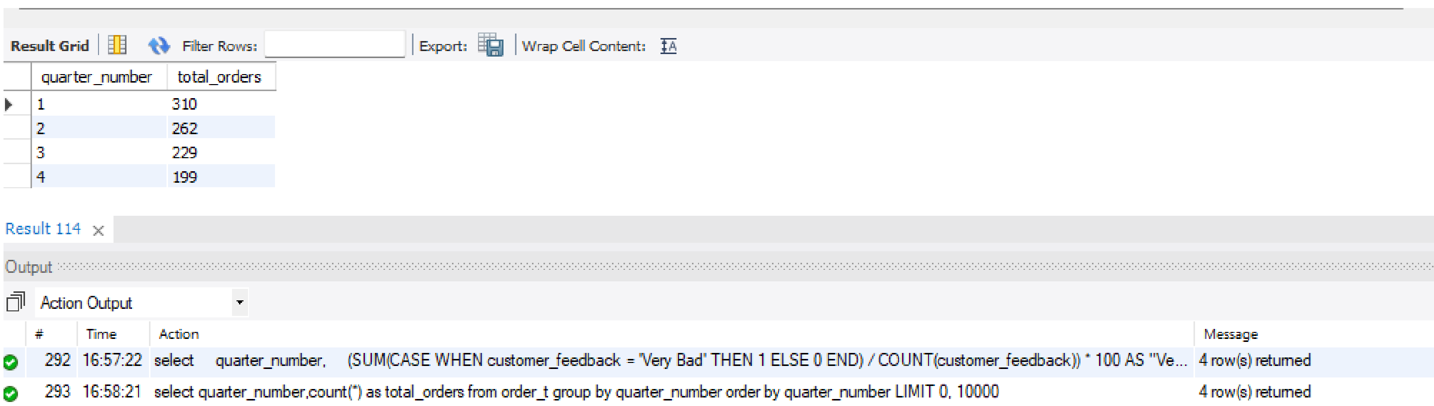
- The steady shift from positive to negative sentiment suggests worsening service quality or customer experience over time.
- Immediate action is needed to identify and address key pain points, especially in after-sales service and shipping processes, to rebuild customer trust.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
select quarter_number,count(*) as total_orders from order_t group by
quarter_number order by quarter_number;
```

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	total_orders
1	310
2	262
3	229
4	199

Result 114

Output

Action Output

	#	Time	Action	Message
✔	292	16:57:22	select quarter_number, (SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(customer_feedback)) * 100 AS "Ve...	4 row(s) returned
✔	293	16:58:21	select quarter_number,count(*) as total_orders from order_t group by quarter_number order by quarter_number LIMIT 0, 10000	4 row(s) returned

Observations and Insights:

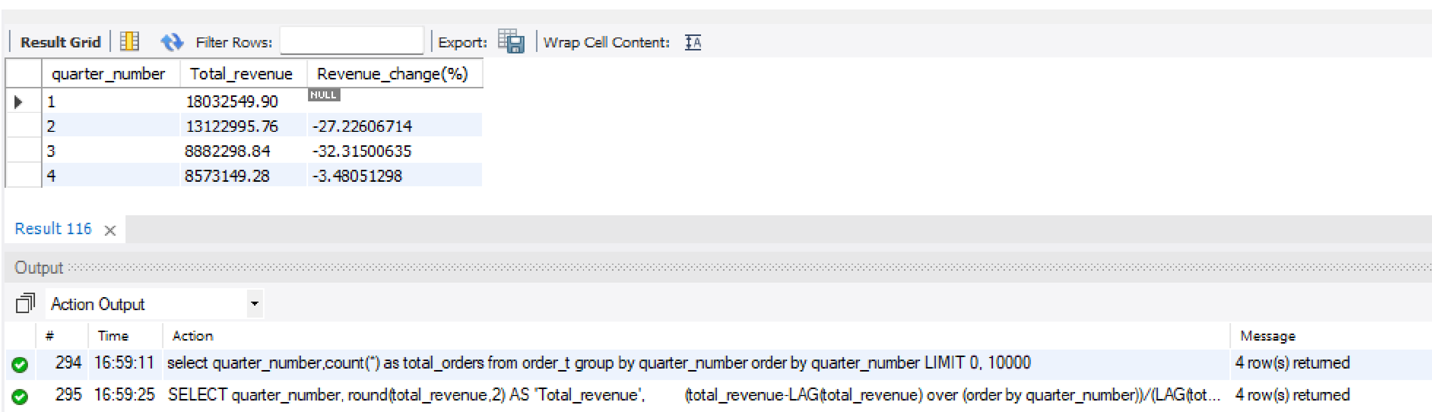
- The total number of orders shows a continuous decline across the quarters from 310 in Q1 to 199 in Q4, indicating a downward trend in customer activity.
- Overall, the company experienced a 35.8% decrease in total orders from the beginning to the end of the year.
- This steady decline suggests weakening customer interest or satisfaction, likely linked to the rising dissatisfaction trends seen in customer feedback.
- Immediate strategic interventions are needed, such as improved service quality, promotional offers, or enhanced post-sales support. It's very important to reverse the falling order trend.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
select quarter_number,
           round(total_revenue,2) AS 'Total_revenue',
        (total_revenue-LAG(total_revenue) over (order by
quarter_number))/(LAG(total_revenue) over (order by quarter_number))*100 AS
'Revenue_change(%)'
      from (   select quarter_number,
         SUM(vehicle_price*quantity*(1-discount)) AS total_revenue
      from order_t
      group by quarter_number
      ) revenue_tb;
```

Output:



quarter_number	Total_revenue	Revenue_change(%)
1	18032549.90	NULL
2	13122995.76	-27.22606714
3	8882298.84	-32.31500635
4	8573149.28	-3.48051298

#	Time	Action	Message
294	16:59:11	select quarter_number,count(*) as total_orders from order_t group by quarter_number order by quarter_number LIMIT 0, 10000	4 row(s) returned
295	16:59:25	SELECT quarter_number, round(total_revenue,2) AS 'Total_revenue', (total_revenue-LAG(total_revenue) over (order by quarter_number))/(LAG(tot...	4 row(s) returned

Observations and Insights:

- The company's net revenue has been declining every quarter, falling from 18.03M in Q1 to 8.57M in Q4, showing a continuous downward trend.
- The largest drop occurred between Q2 and Q3, with revenue decreasing by 32.3%, following an earlier 27.2% decline from Q1 to Q2.
- By Q4, the decline slowed to 3.5%, but revenue still remained significantly lower than the start of the year (a 52.5% overall drop from Q1 to Q4).
- The consistent decrease in revenue likely correlates with declining order volumes and worsening customer satisfaction, as observed in previous analyses.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
select quarter_number,

 round( sum(vehicle_price*quantity*(1-discount)),2)AS total_revenue,

count(*) as total_orders from order_t GROUP BY quarter_number order by
quarter_number;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	total_revenue	total_orders
1	18032549.90	310
2	13122995.76	262
3	8882298.84	229
4	8573149.28	199

Result 119

Output

Action Output

#	Time	Action	Message
297	17:00:26	SELECT quarter_number, round(total_revenue,2) AS 'Total_revenue', (total_revenue-LAG(total_revenue) over (order by quarter_number))/(LAG(tot...	4 row(s) returned
298	17:00:34	select quarter_number, round(sum(vehicle_price*quantity*(1-discount)),2)AS total_revenue, count(*) as total_orders from order_t GROUP BY quarter_n...	4 row(s) returned

Observations and Insights:

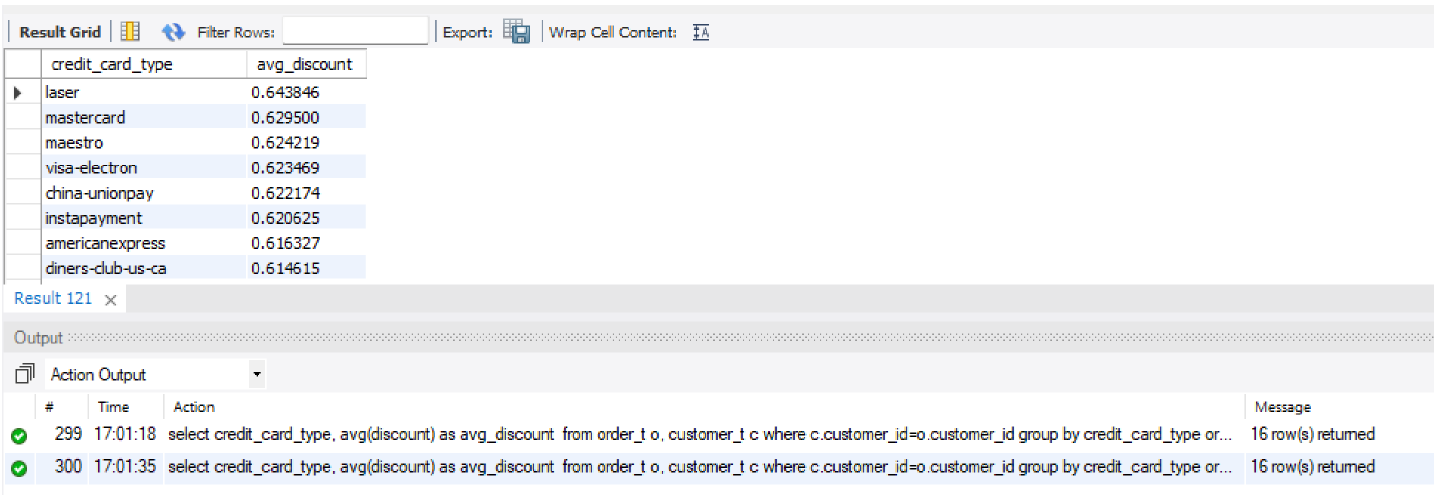
- Both net revenue and total orders show a steady decline across quarters — from 18.03M and 310 orders in Q1 to 8.57M and 199 orders in Q4.
- The parallel downward trend in both metrics points to decreasing customer retention and satisfaction, consistent with the feedback deterioration observed earlier.
- The company should investigate service quality, pricing, and customer experience issues to reverse this declining trajectory and stabilize future revenues.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
select credit_card_type, avg(discount) as avg_discount

from order_t o, customer_t c where c.customer_id=o.customer_id group by
credit_card_type order by avg_discount desc ;
```

Output:



credit_card_type	avg_discount
laser	0.643846
mastercard	0.629500
maestro	0.624219
visa-electron	0.623469
china-unionpay	0.622174
instapayment	0.620625
americanexpress	0.616327
diners-club-us-ca	0.614615

#	Time	Action	Message
299	17:01:18	select credit_card_type, avg(discount) as avg_discount from order_t o, customer_t c where c.customer_id=o.customer_id group by credit_card_type or...	16 row(s) returned
300	17:01:35	select credit_card_type, avg(discount) as avg_discount from order_t o, customer_t c where c.customer_id=o.customer_id group by credit_card_type or...	16 row(s) returned

Observations and Insights:

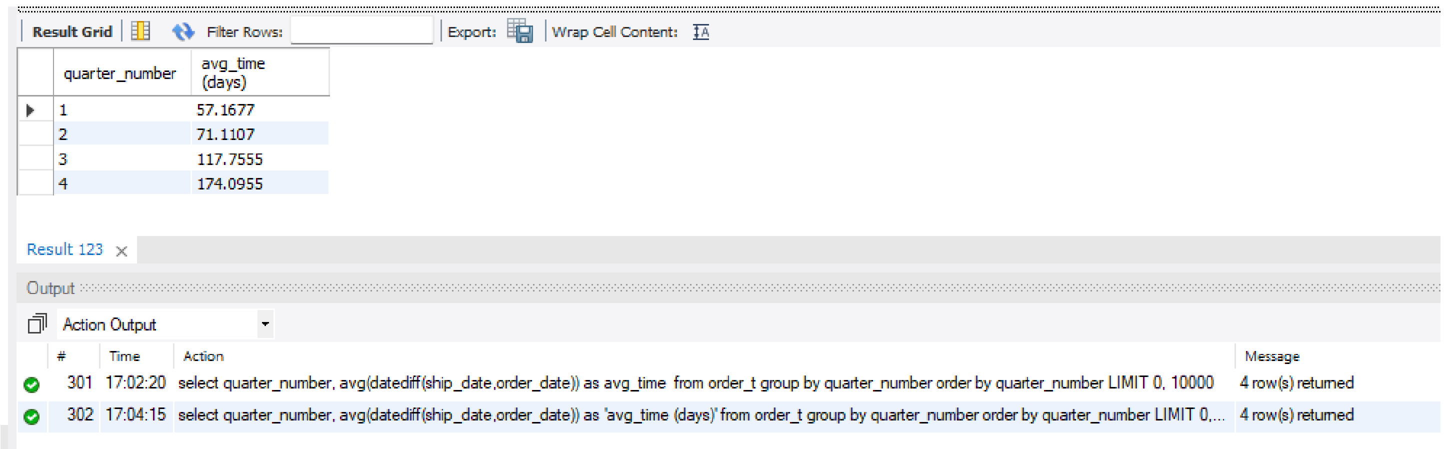
- The average discount offered across credit card types is relatively close, ranging from ~0.58 to 0.64, indicating a fairly uniform discount policy.
- 'Laser' cardholders receive the highest average discount (0.6438), while 'Diners Club International' users get the lowest (0.5840).
- To drive more sales, the company could consider targeted or card-specific promotional discounts to attract higher-spending customer segments.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
select quarter_number, avg(datediff(ship_date,order_date)) as 'avg_time (days)'
from order_t group by quarter_number order by quarter_number;
```

Output:



quarter_number	avg_time (days)
1	57.1677
2	71.1107
3	117.7555
4	174.0955

Observations and Insights:

- The average shipping time has increased sharply each quarter, from 57 days in Q1 to 174 days in Q4, and the average delay more than tripled over the year, largest jump occurring between Q2 (71 days) and Q3 (118 days), indicating a significant slowdown in delivery efficiency.
- By Q4, the shipping process had deteriorated severely, taking over 4 months on average to deliver an order.
- This steady rise in delivery time likely contributed to increased customer dissatisfaction and declining order volumes observed in previous analyses.
- The company must review logistics, vendor coordination, and shipping partner performance to restore timely deliveries and rebuild customer trust.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48,610,993.78	1000	994	3.0657
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8,573,149.28	199	98 approx	21.50%

Business Recommendations

1. Improve Delivery and Logistics Efficiency
   - The average shipping time has increased from 57 to 174 days, which is extremely high and likely the primary driver of poor customer satisfaction.
   - Partner with reliable logistics providers, implement performance-based contracts, and use real-time shipment tracking to ensure timely deliveries.
2. Strengthen Customer Experience and After-Sales Support
   - Only 21.5% of customers gave “Good” feedback, and the average rating (3.07/5) indicates moderate to low satisfaction.
   - Set up a dedicated customer service team to handle complaints faster and improve communication throughout the purchase-to-delivery process.
3. Boost Customer Retention and Loyalty
   - Orders have fallen from 310 in Q1 to 199 in Q4, a 36% decline; this drop is directly affecting revenue.
   - Introduce loyalty rewards, referral bonuses, and seasonal offers to encourage repeat purchases and attract new buyers.
4. Focus on Service Quality and Brand Reputation
   - The company’s revenue dropped 52.5% from Q1 to Q4, and feedback shows increasing dissatisfaction.
   - Launch customer feedback-driven improvement programs, use post-sale surveys to identify pain points, and showcase visible improvements to rebuild trust.
5. Data-Driven Operational Monitoring
   - Establish a quarterly performance dashboard tracking key KPIs of orders, revenue, feedback, and shipping time to detect early warning signs.
   - Use predictive analytics to forecast demand, optimize inventory, and allocate resources efficiently.

Overall Insight:

The business is experiencing a systemic decline in performance across all key areas: orders, revenue, delivery, and satisfaction. To recover, operational efficiency, customer-centric service, and data-driven decision-making must be prioritized immediately.

Thank You

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